TIME SERIES ANALYSIS

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3. Autoregressive Process

Autoregressive processes

- Assume $\{e_t\}$ is purely random with mean zero and s.d. σ_e
- Then the autoregressive process of order p or AR(p) process is

$$y_t = \varphi_1 y_t + \varphi_2 y_{t-2} + \cdots + \varphi_p y_{t-p} + e_t$$

Autoregressive processes

- The first order autoregression is

$$y_t = \varphi y_t + e_t$$

- Provided $|\varphi| < 1$ it may be written as an infinite order MA process

- Using the backshift operator we have

$$(1 - \varphi B)y_t = e_t$$

Autoregressive processes

- From the previous equation we have

$$y_t = \frac{e_t}{(1 - \varphi B)}$$

$$y_t = (1 + \varphi B + \varphi^2 B^2 + \cdots)e_t$$

$$y_t = e_t + \varphi e_{t-1} + \varphi^2 e_{t-2} + \cdots$$

Autoregressive processes

- Then $E(y_t) = 0$, and if $|\varphi| < 1$

$$var(y_t) = \sigma_e^2 = \sigma_y^2 = \sigma^2/(1-\varphi^2)$$

$$\gamma_k = \varphi^k \sigma_e^2/(1-\varphi^2)$$

$$\rho_k = \varphi^k$$

Autoregressive processes

- The AR(p) process can be written as

$$(1+\varphi_1 B+\varphi_2 B^2+\cdots+\varphi_p B^p)y_t = e_t$$

$$or$$

$$y_t = e_t/(1+\varphi_1 B+\varphi_2 B^2+\cdots+\varphi_p B^p) = f(B)e_t$$

Autoregressive processes

- This is for

$$f(B) = (1 + \varphi_1 B + \varphi_2 B^2 + \cdots + \varphi_p B^p)^{-1}$$

$$f(B) = (1 + \beta_1 B + \beta_2 B^2 + \cdots + \beta_p B^p)$$

for some $\beta_1, \beta_2, \ldots.$

This gives y_t as an infinite MA process, so it has mean zero

Autoregressive processes

- Conditions are needed to ensure that various series converge, and hence that the variance exists, and the autocovariance can be defined

- Essentially these are requirements that the β_i become small quickly enough, for large i

Autoregressive processes

An equivalent way of expressing this is that the roots of the equation

$$f(B) = (1 + \varphi_1 B + \varphi_2 B^2 + \cdots + \varphi_p B^p)$$

must lie outside the unit circle.

AR: Stationarity

- Suppose y_t follows an AR(1) process without drift.
- Is y_t stationarity?
- Note that

$$y_t = \varphi_1 y_{t-1} + e_t$$

$$y_t = \varphi_1(\varphi_1 y_{t-2} + e_{t-1}) + e_t$$

$$y_t = e_t + \varphi_1 e_{t-1} + \varphi_1^2 e_{t-2} + \varphi_1^3 e_{t-3} + \cdots + \varphi_1^t y_0$$

Stationarity

- Without loss of generality, assume that $y_0 = 0$. Then $E(y_t) = 0$.

- Assuming that t is large, i.e., the process started a long time ago, then $\mathrm{var}(y_t) \quad \frac{\quad ^2}{(1 \quad _1^2)}$, provided that $| \quad _1 | \quad 1.$ It can also be shown that provided that the same condition is satisfied, $\mathrm{cov}(y_t\, y_{t \quad s}) \quad \frac{\quad _1^s \quad ^2}{(1 \quad _1^2)} \quad _1^s \ \mathrm{var}(y_t)$

Stationarity

- Suppose the model is an AR(2) without drift, i.e., $y_t \quad {}_1 y_{t \quad 1} \quad {}_2 y_{t \quad 2} \quad {}_t$
- It can be shown that for y_t to be stationary,
- The key point is that AR processes are not stationary unless appropriate prior conditions are imposed on the parameters.

$$ {}_1 \quad {}_2 \quad 1, \quad {}_2 \quad {}_1 \quad 1 \text{ and } | \quad {}_2 \, | \quad 1 $$

AR(1) Process:

$$y_t = \phi y_{t-1} + u_t$$

ACVF: $\gamma_k = \sigma_y^2 \phi^{|k|};\ (k = 0, \pm 1, \pm 2, \dots)$

Power spectrum function is then

$$f(\omega) = \frac{\sigma_y^2}{2\pi}\left[1 + \sum_{k=1}^{\infty} \phi^k e^{-i\omega k} + \sum_{k=1}^{\infty} \phi^k e^{i\omega k}\right]$$

$$= \frac{\sigma_u^2}{2\pi(1 - 2\phi Cos\omega + \phi^2)}; -\pi \le \omega \le \pi$$

$(\sigma_u^2 = (1 - \phi^2)\sigma_Y^2)$

$$f(\omega) = \frac{\sigma_y^2}{2\pi}\left[1 + \frac{\phi e^{-i\omega}}{1 - \phi e^{-i\omega}} + \frac{\phi e^{i\omega}}{1 - \phi e^{i\omega}}\right]$$

$$= \frac{1 - 2\phi \cos\omega + \phi^2 + 2\phi \cos\omega - 2\phi^2}{1 - 2\phi \cos\omega + \phi^2}$$

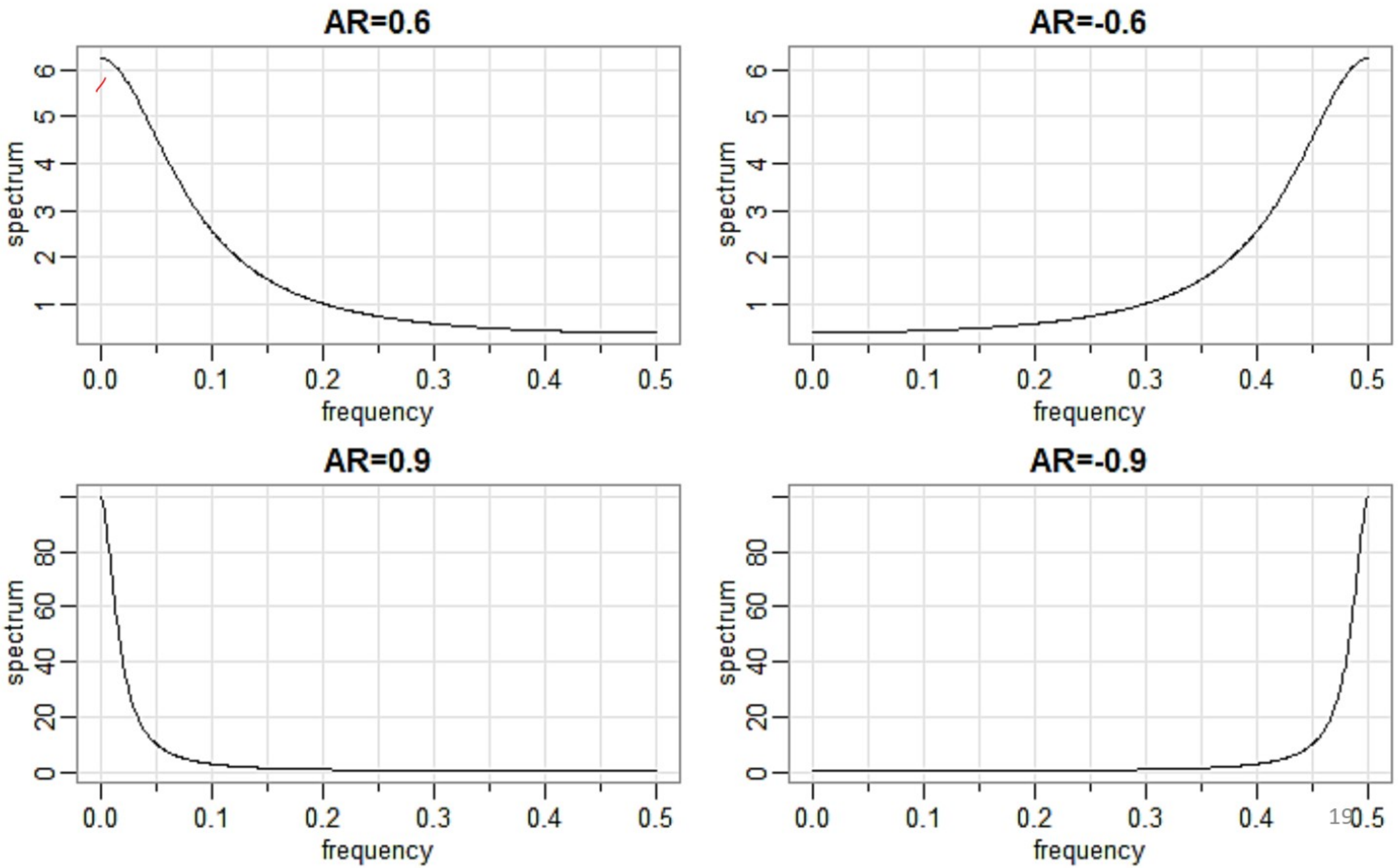
Normalized Power spectrum function is

$$f^*(\omega) = \frac{1}{2\pi}\left[1 + \sum_{k=1}^{\infty} \phi^k e^{-i\omega k} + \sum_{k=1}^{\infty} \phi^k e^{i\omega k}\right]$$

$$= \frac{\sigma_u^2(1-\phi^2)}{2\pi(1-2\phi\cos\omega+\phi^2)}$$

When $\phi > 0$, power is concentrated at low frequencies.

When $\phi < 0$, power is concentrated at high frequencies.



AR(1) Process: p=1

$$y_t = \phi_1 y_{t-1} + \delta + u_t$$

Given y_{t-1}, y_t becomes

independent of $y_{t-2}, y_{t-3}, \ldots.$

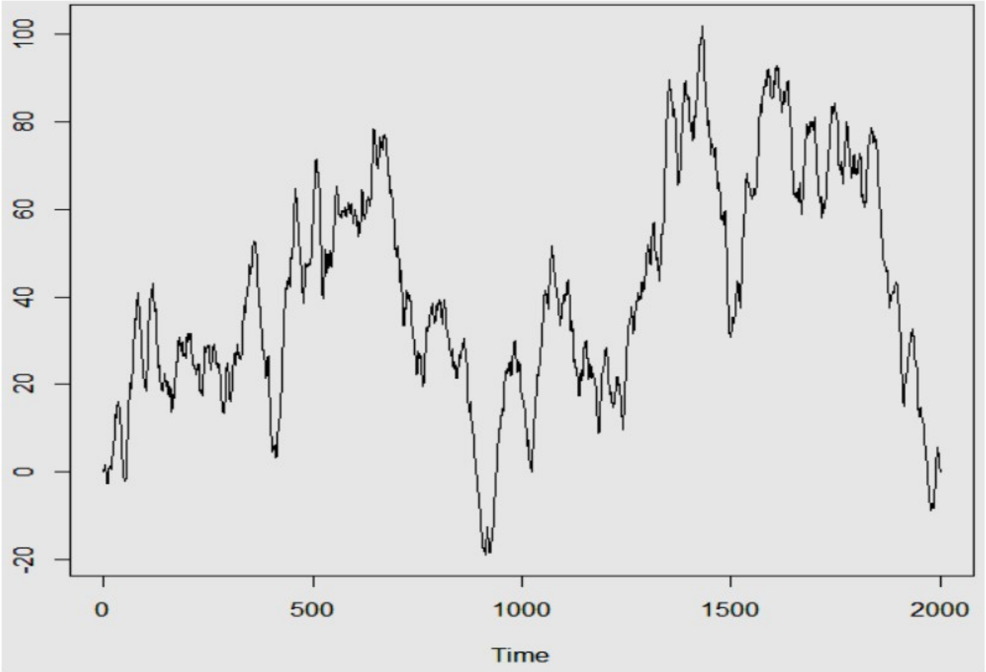
(Markovian property)

The process is also called Markov

process

Mean of the process: $\mu = \frac{\delta}{1-\phi_1}$

Simulated data from AR(1) process
$\delta = 0$, $\phi_1 = 0.7$, n=2000



Take $\delta = 0$, so that $\mu = 0$. Assume that the process is variance stationary.

$$\gamma_0 = Var(y_t) = Var(y_{t-1}) = \cdots = \sigma_y^2 \quad \forall t$$

Then

$$\sigma_y^2 = E[\phi_1 y_{t-1} + u_t]^2 \quad = \phi_1^2 E y_{t-1}^2 + E u_t^2 + 2\phi_1 \underline{E(y_{t-1} u_t)}$$

$$= \phi_1^2 \sigma_y^2 + \sigma_u^2$$

$$(1 - \phi_1^2)\sigma_y^2 = \sigma_u^2$$

$E[u_t y_{t-1}] = 0$, (y_{t-1} depends only on $u_{t-1}, u_{t-2}, \ldots$)

$$\sigma_y^2 = \frac{\sigma_u^2}{1-\phi_1^2} = \gamma_0$$

ACVF and ACF:

$$\gamma_1 = E(y_t y_{t-1}) = \phi_1 \gamma_0 = \frac{\phi_1 \sigma_u^2}{1-\phi_1^2}.$$

$$E(y_t\, y_{t-1}) = E\left[(\phi_1 y_{t-1} + u_t)\, y_{t-1}\right]$$

$$= \phi_1 \gamma_0$$

Further

$y_{t-k} \rightarrow u_{t-k}, u_{t-k-1}, \ldots$

$E(y_{t-k}u_t) = 0, \forall k = 1,2, \ldots$

$$\gamma_k = E[y_t y_{t-k}] = E[(\phi_1 y_{t-1} + u_t)y_{t-k}] \quad = \phi_1 \gamma_{k-1} + 0$$

$$= \phi_1\gamma_{k-1} = \phi_1\phi_1\gamma_{k-2} = \phi_1^2\gamma_{k-2} = \cdots = \phi_1^k\gamma_0$$

$\gamma_{k-1} = \phi_1 \gamma_{k-2}$

$$= \frac{\phi_1^k \sigma_u^2}{1 - \phi_1^2}$$

ACF of the AR(1) process

$E(y_{t-1} y_{t-k})$

$= \gamma_{(t-1)-(t-k)}$

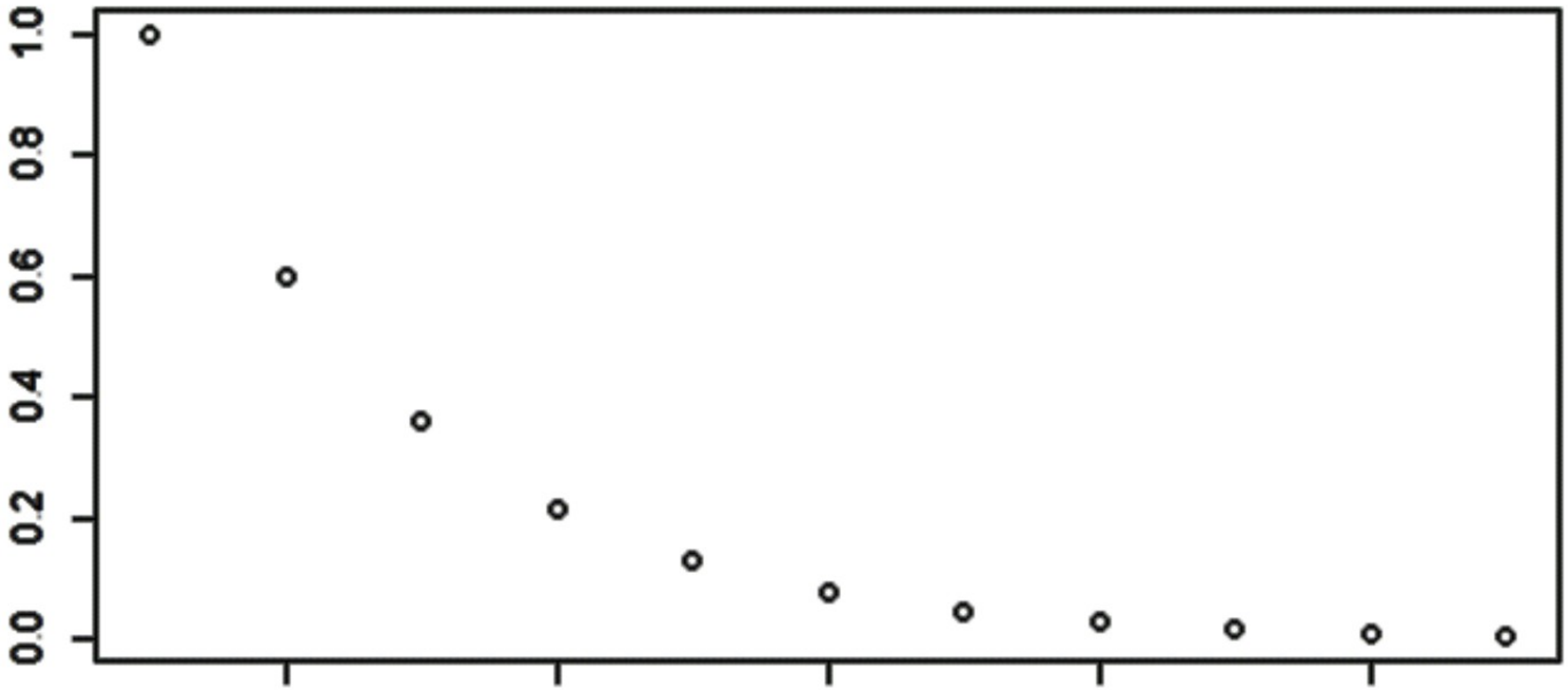
$= \gamma_{k-1}$

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \phi_1^k.$$

Since $|\phi_1| < 1$, ρ_k declines geometrically.

ACF of the AR(1) process: $y_t = 0.6y_{t-1} + u_t$

ACF: $\rho_k = (0.6)^k$



Example: AR(2) process with $\delta = 0$

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + u_t$$

We have

$$\gamma_0 = E(y_t^2)$$

$$= E\{(\phi_1 y_{t-1} + \phi_2 y_{t-2} + u_t) y_t\}$$

$$= \phi_1 \gamma_1 + \phi_2 \gamma_2 + \sigma_u^2 \quad (E(u_t y_t) = \sigma_u^2)$$

$$\gamma_1 = \phi_1 \gamma_0 + \phi_2 \gamma_1$$

$$\gamma_2 = \phi_1 \gamma_1 + \phi_2 \gamma_0.$$

$p = 2$

$E\, u_t y_t$

$= E[u_t(\phi_1 y_{t-1} + \phi_2 y_{t-2} + u_t)]$

$= \sigma_u^2$

$E(y_t\, y_{t-1})$

$= E(\phi_1 y_{t-1} + \phi_2 y_{t-2} + u_t) \times y_{t-1}$

$= \phi_1 \gamma_0 + \phi_2 \gamma_1$

$E(y_t\, y_{t-2})$

$= E[(\phi_1 y_{t-1} + \phi_2 y_{t-2} + u_t) y_{t-2}]$

$= \phi_1 \gamma_1 + \phi_2 \gamma_0 + 0$

Hence, we obtain

$$\gamma_1 = \frac{\phi_1}{1-\phi_2}\gamma_0,$$

$$\gamma_2 = \left(\frac{\phi_1^2}{1-\phi_2} + \phi_2\right)\gamma_0,$$

$$\gamma_0 = \frac{1-\phi_2}{1+\phi_2}\frac{\sigma_u^2}{(1-\phi_2)^2 - \phi_1^2}$$

For MA processes the correlogram vanishes after a certain point.

For AR process it declines but never vanishes.

Yule-walker Equations:

- Relate the autoregressive model parameters to the autocovarianc
- Provide a straightforward method for parameters estimation

AR(p) process

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + \cdots + \phi_p y_{t-p} + u_t$$

Multiplying by y_{t-k} and taking expectation, we obtain

$$E[y_t y_{t-k}]$$

$$= \phi_1 E[y_{t-1} y_{t-k}] + \phi_2 E[y_{t-2} y_{t-k}] + \cdots + \phi_p E[y_{t-p} y_{t-k}]$$

$$+ E[u_t y_{t-k}] \qquad (1)$$

Substituting $k = 0,1,2, \ldots, p$ in (1)

$$E[y_t^2] = \phi_1 E[y_{t-1} y_t] + \phi_2 E[y_{t-2} y_t] + \cdots + \phi_p E[y_{t-p} y_t] + E[u_t y_t]$$

$$E[y_t y_{t-1}]$$

$$= \phi_1 E[y_{t-1}^2] + \phi_2 E[y_{t-2} y_{t-1}] + \cdots + \phi_p E[y_{t-p} y_{t-1}] + E[u_t y_{t-1}]$$

$$E[y_t y_{t-2}]$$

$$= \phi_1 E[y_{t-1} y_{t-2}] + \phi_2 E[y_{t-2}^2] + \cdots + \phi_p E[y_{t-p} y_{t-2}] + E[u_t y_{t-2}]$$

$$\vdots$$

$$E[y_t y_{t-p}]$$

$$= \phi_1 E[y_{t-1} y_{t-p}] + \phi_2 E[y_{t-2} y_{t-p}] + \cdots + \phi_p E[y_{t-p}^2] + E[u_t y_{t-p}] \quad (2)$$

We observe that

$$E[y_t u_t]$$

$$= E\left[\left(\phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + \cdots + \phi_p y_{t-p} + u_t\right) u_t\right]$$

$$= E[u_t^2] = \sigma_u^2$$

$$E[y_{t-k} u_t] = 0 \ \forall \ k = 1,2, \dots, p.$$

Hence, set of equations (2) reduces to

$$\left.\begin{aligned} \gamma_0 &= \phi_1\gamma_1 + \phi_2\gamma_2 + \cdots + \phi_p\gamma_p + \sigma_u^2 \\ \gamma_1 &= \phi_1\gamma_0 + \phi_2\gamma_1 + \cdots + \phi_p\gamma_{p-1} \\ &\vdots \\ \gamma_p &= \phi_1\gamma_{p-1} + \phi_2\gamma_{p-2} + \cdots + \phi_p\gamma_0 \end{aligned}\right\} \qquad (3)$$

Dividing each of $p+1$ equations of (3) by γ_0 gives the following $p+1$ equations:

$$\left.\begin{array}{l} \rho_0 = \phi_1\rho_1 + \phi_2\rho_2 + \cdots + \phi_p\rho_p + \sigma_u^2/\gamma_0 \\ \rho_1 = \phi_1\rho_0 + \phi_2\rho_1 + \cdots + \phi_p\rho_{p-1} \\ \vdots \\ \rho_p = \phi_1\rho_{p-1} + \phi_2\rho_{p-2} + \cdots + \phi_p\rho_0 \end{array}\right\} \quad (4)$$

Here $\rho_0 = 1$.

The equations in (4) jointly determine the p values of ACF and called *Yule-Walker equations*.

Let us write

$$\rho_{(p)} = (\rho_1 \;\; \rho_2 \cdots \rho_p)';$$

$p \times 1$

$$\phi = (\phi_1 \;\; \phi_2 \;\; \cdots \phi_p)';$$

$$P_{(p)} = \begin{pmatrix} 1 & \rho_1 & \cdots & \rho_{p-1} \\ \rho_1 & 1 & & \rho_{p-2} \\ \vdots & & \ddots & \vdots \\ \rho_{p-1} & \rho_{p-2} & \cdots & 1 \end{pmatrix}.$$

We can write Yule-Walker equations as

$$1 = \rho'_{(p)}\phi + \frac{\sigma_u^2}{\gamma_0}$$

$$\rho_{(p)} = P_{(p)}\phi.$$

$$\rho_0 = \phi_1\rho_1 + \phi_2\rho_2 + \cdots + \phi_p\rho_p$$

$$+\sigma_u^2/\gamma_0$$

$$\rho_1 = \phi_1\rho_0 + \phi_2\rho_1 + \cdots + \phi_p\rho_{p-1}$$

$$\vdots$$

$$\rho_p = \phi_1\rho_{p-1} + \phi_2\rho_{p-2} + \cdots + \phi_p\rho_0$$

$$\rho_0 = 1$$

$$\phi = P_{(p)}^{-1}\rho_{(p)}$$

Partial ACF

- We learnt earlier that correlation-based measures suffer from confounding, i.e., the common influence of a third extraneous variable can cause two variables to appear as correlated.
- The correlation between two observations of a series (or two different series) is most likely to suffer from confounding because the intermediate samples can introduce an apparent correlation due to propagated effects. While this phenomenon holds for any random process, it becomes particularly important for auto-regressive processes.
- We understand the issue of confounding in ACF by revisiting the ACF of an AR(1) process.

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ACF of an AR(1) process

Recall the ACF of an AR(1) process v[k]= —d1v[k — 1] + e[k]

$$\rho_{vv}[l] = (-d_1)^{|l|}$$

The ACF suggests that v[k] and v[k — l] are correlated whereas the governing difference equation for the process clearly shows that only two successive samples v[k] and v[k — 1] directly influence each other.

ACF of an AR(1) process

- Q: What is the cause of this apparent correlation between samples separated by las
L > 1?
- A: The cause for this apparent correlation is the propagated effect. For instance, the
difference equation of the process can be re-written as:

$$v[k] = -d_1(-d_1 v[k-2] + e[k-1]) + e[k] = d_1^2 v[k-2] - d_1 e[k-1] + e[k]$$

Thus v[k − 2] appears to influence v[k] indirectly through v[k − 1]. The same argument can be extended to explain correlation at other lags as well.

How do we ensure ACF measures direct correlations only?

Conditioned ACF: Partial ACF

- To measure the direct correlation between $v[k - l]$ and $v[k]$ we should account for the possible propagated effects of the intermediate variables $\{v[k - l + 1, \cdots, v[k - 1]\}$.

- The procedure is illustrated for $l = 2$. The idea is to remove the presence of $v[k - 1]$ in both $v[k]$ and $v[k - 2]$ followed by a correlation between the respective residuals. The resulting correlation is known as partial auto-correlation function (PACF)

Partial ACF

- Partial ACF is analogous to “partial derivative” where only the effects w.r.t. a specific variable are evaluated.
- As we learnt earlier, computing partial correlation (or any other measure) is known as conditioning in signal processing
- The partial ACF measures direct correlation whereas the ACF measures total correlation

Procedure to compute PACF

- Obtain the best predictor for v[k] using v[k − 1]. Denote the associated residuals by $\eta[k]$

$$\hat{v}[k|v[k-1]] = \alpha_1 v[k-1]; \qquad \eta[k] = v[k] - \alpha_1^\star v[k-1]$$

- Obtain the best "predictor" for v[k − 2] using v[k − 1]. Denote the associated residuals by $\eta[k-2]$.

$$\hat{v}[k-2|v[k-1]] = \beta_1 v[k-1]; \qquad \eta[k-2] = v[k-2] - \beta_1^\star v[k-1]$$

- Compute $\phi_{vv}[2] = \text{corr}(\eta[k], \eta[k-2])$ to obtain the PACF of the series v[k] at lag 2

Procedure to compute PACF . . . contd.

- The optimal estimates of α_1 and β_1 are obtained in such a way that η[k] and η[k −2] do not contain any (linear) effects of v[k − 1], i.e.,

$$\text{corr}(\eta[k], v[k-1]) = 0 \qquad \text{corr}(\eta[k-2], v[k-1]) = 0$$

- These are also the conditions of optimality for the least squares technique. Thus, α_1^* and β_1^* are the LS estimates.

$$\boxed{\alpha_1^\star = \rho_{vv}[1]} \qquad \boxed{\beta_1^\star = \rho_{vv}[1]}$$

General procedure

- Obtain the best predictors for v[k] and v[k − l] using {v[k − 1], v[k − 2], · · · , v[k − l + 1]}. Denote the associated residuals by η[k] and η[k − l] respectively.

$$\eta[k] = v[k] - \sum_{j=1}^{l-1} \alpha_j^\star v[k-j] \qquad \eta[k-l] = v[k-l] - \sum_{j=1}^{l-1} \beta_j^\star v[k-l+j]$$

- Compute $\phi_{vv}[l] = corr(n[k], n[k-l])$ to obtain the PACF at lag l.

Alternative procedure

- The PACF coefficient at any lag p, $\phi_{vv}[p]$ can be shown to be the last coefficient of an AR(p) model fit to the series v[k].

- Fit an AR(l) model at each lag l.
- Determine the PACF at any lag l as the last coefficient of that model.
- A recursive algorithm due to Durbin and Levinson is used in practice to compute $\phi_{vv}[p]$using the coefficient at $l = p - 1$ and the ACF coefficients.

Remarks

- The "prediction" of $v[k-l]$ using future values is known as backcasting
- The PACF at lag $l=0$ is not defined. However, to be consistent with ACF, PACF at lag $l=0$ maybe set to unity.

PACF of an AR(1) process

- Problem: Find the PACF of an AR(1) process: v[k] = $-d_1$v[k − 1] + e[k] at lags l = 1, 2.
- Solution: The PACF coefficient at lag l = 1 is the ACF at lag l = 1 itself since there is no intermediate variable. The direct correlation at lag l = 1 is the same as total correlation.

- Thus

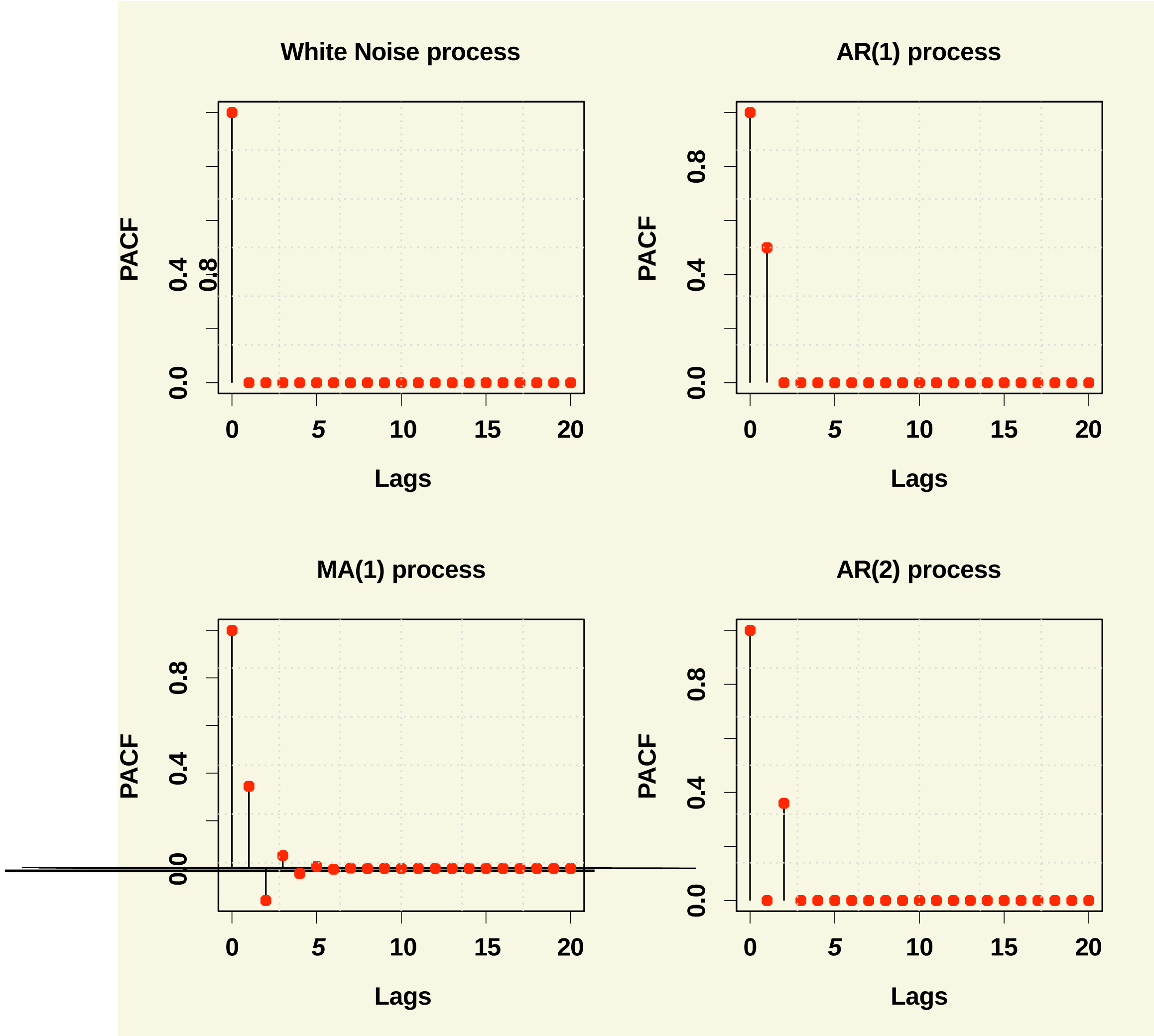
$$\phi_{vv}[1] = \rho_{vv}[1] = (-d_1)^{|l|}$$

PACF of an AR(1) process

- To compute the PACF at lag l = 2, recall from the procedure

$$\begin{aligned}
\phi_{vv}[2] &= \text{corr}(v[k] - \alpha_1^\star v[k-1], v[k-2] - \beta_1^\star v[k-1]) \\
&= \frac{\text{cov}(v[k] - \rho_{vv}[1]v[k-1], v[k-2] - \rho_{vv}[1]v[k-1])}{\sqrt{\text{var}(v[k] - \rho_{vv}[1]v[k-1])\text{var}(v[k-2] - \rho_{vv}[1]v[k-1])}} \\
&= \frac{\rho[2] - \rho[1]}{1 - \rho[1]^2}
\end{aligned}$$

Theoretical PACF: Examples



Remarks

- The PACF of a WN process is zero at all lags (like the ACF)
- PACF of an MA(1) process dies down exponentially (somewhat analogous to the behaviour of ACF for an AR(1) process).
- The notion of PACF can be extended to handle negative lags as well. For stationary processes, PACF is symmetric like the ACF

Consider an AR(1) process, $x_t = \phi x_{t-1} + w_t$. Note that $x_{t-1} = \phi x_{t-2} + w_{t-1}$, substituting back

$$x_t = \phi^2 x_{t-2} + \phi w_{t-1} + w_t.$$

Again, $x_{t-2} = \phi x_{t-3} + w_{t-2}$, substituting back

$$x_t = \phi^3 x_{t-3} + \phi^2 w_{t-2} + \phi w_{t-1} + w_t.$$

Continuing this process, we could rewrite x_t as

$$x_t = w_t + \phi w_{t-1} + \phi^2 w_{t-2} + \phi^3 w_{t-3} + ...$$

(note that x_t involves $\{w_t, w_{t-1}, w_{t-2}, w_{t-3}, ...\}$)

To formally define the PACF for mean-zero stationary time series, let $\hat{x}_{t+h}$, for $h \geq 2$, denote the regression of x_{t+h} on $\{x_{t+h-1}, x_{t+h-2}, ..., x_{t+1}\}$ which we write as

$$\hat{x}_{t+h} = \beta_1 x_{t+h-1} + \beta_2 x_{t+h-2} + ... + \beta_{h-1} x_{t+1}.$$

NO intercept is needed because the mean of x_t is zero.

In addition, let $\hat{x}_t$ denote the regression of x_t on $\{x_{t+1}, x_{t+2}, ..., x_{t+h-1}\}$, then

$$\hat{x}_t = \beta_1 x_{t+1} + \beta_2 x_{t+2} + ... + \beta_{h-1} x_{t+h-1}.$$

The partial autocorrelation function (PACF) of a stationary process, x_t, denoted ϕ_h^h (or ϕ_{hh}) , for $h = 1, 2, ...$ is

$$\phi_{11} = corr(x_{t+1}, x_t) = \rho(1)$$

and

$$\phi_{hh} = corr(x_{t+h} - \hat{x}_{t+h}, x_t - \hat{x}_t), \quad h \geq 2.$$

Suppose we suspect that $p = 2$; that is, we suspect that we are dealing with an AR(2) process. The Yule-Walker equations are:

$E[Y_t Y_{t-1}] = a_1 E[Y_{t-1}^2] + a_2 E[Y_{t-2} Y_{t-1}] + E[W_t Y_{t-1}]$

$\gamma(1) = a_1 \gamma(0) + a_2 \gamma(1)$

$\frac{\gamma(1)}{\gamma(0)} = a_1 + a_2 \frac{\gamma(1)}{\gamma(0)}$

$$\rho(1) = a_1 + a_2 \rho(1) \tag{1}$$

$$E[Y_t Y_{t-2}] = a_1 E[Y_{t-2}^2] + a_2 E[Y_{t-2} Y_{t-2}] + E[W_t Y_{t-2}]$$
$$\gamma(2) = a_1 \gamma(1) + a_2 \gamma(0)$$
$$\frac{\gamma(2)}{\gamma(0)} = a_1 \frac{\gamma(1)}{\gamma(0)} + a_2$$

$$\rho(2) = a_1 \rho(1) + a_2$$

Or, in matrix form,

$$\begin{pmatrix} \rho(1) \\ \rho(2) \end{pmatrix} = \begin{pmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

The second-order partial autocorrelation coefficient is a_2, written ϕ_{22}, which can be found using Cramer's rule,

$$a_2 = \phi_{22} = \frac{det\begin{pmatrix} 1 & \rho(1) \\ \rho(1) & \rho(2) \end{pmatrix}}{det\begin{pmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{pmatrix}} = \frac{\rho(2) - \rho^2(1)}{1 - \rho^2(1)}$$

LEVINSON DURBIN ALGORITHM

For a given lag k, it can be shown that the ϕ_{kk} satisfy the Yule-Walker equations:

$$\rho_j = \phi_{k1}\rho_{j-1} + \phi_{k2}\rho_{j-2} + \phi_{k3}\rho_{j-3} + \cdots + \phi_{kk}\rho_{j-k}$$

for $j = 1, 2, \cdots, k$.
Here we are treating $\rho_1, \rho_2, \cdots, \rho_k$ as given and wish to solve for $\phi_{k1}, \phi_{k2}, \cdots, \phi_{kk}$.

LEVINSON DURBIN ALGORITHM

Levinson and Durbin gave an efficient method for obtaining the solutions to equations given on previous slide, for either theoretical or sample autocorrelations. They showed that these equations can be solved recursively as follows:

$$\phi_{kk} = \frac{\rho_k - \sum_{j=1}^{k-1} \phi_{k-1,j}\rho_{k-j}}{1 - \sum_{j=1}^{k-1} \phi_{k-1,j}\rho_j}$$

where

$$\phi_{k,j} = \phi_{k-1,j} - \phi_{kk}\phi_{k-1,k-j}$$

for $j = 1, 2, \cdots, k-1$

Another way to introduce PACF is to consider the following AR models in consecutive orders:

$$
\begin{aligned}
y_t &= \phi_{0,1} + \phi_{1,1} y_{t-1} + w_{1t}, \\
y_t &= \phi_{0,2} + \phi_{1,2} y_{t-1} + \phi_{2,2} y_{t-2} + w_{2t}, \\
y_t &= \phi_{0,3} + \phi_{1,3} y_{t-1} + \phi_{2,3} y_{t-2} + \phi_{3,3} y_{t-3} + w_{3t}, \\
y_t &= \phi_{0,4} + \phi_{1,4} y_{t-1} + \phi_{2,4} y_{t-2} + \phi_{3,4} y_{t-3} + \phi_{4,4} y_{t-3} + w_{4t}, \\
&\vdots
\end{aligned}
$$

where $\phi_{0,j}, \phi_{i,j}$, and $\{w_{jt}\}$ are, respectively, the constant term, the coefficient of y_{t-i}, and the error term of an AR(j) model. These models are in the form of a multiple linear regression and can be estimated by the least-squares method. The estimate $\hat{\phi}_{1,1}$ of the first equation is called the lag-1 sample PACF of y_t. The estimate $\hat{\phi}_{2,2}$ of the second equation is called the lag-2 sample PACF of y_t. The estimate $\hat{\phi}_{3,3}$ of the third equation is called the lag-3 sample PACF of y_t, an so on.

Estimation of Parameters of Autoregressive Processes

$y_1, y_2, \ldots, y_n$: Observed time series

Sample mean: $\bar{y} = \frac{1}{n}\sum_{t=1}^{n} y_t$

Sample ACVF: $c_k = \frac{1}{n-k}\sum_{t=1}^{n-k}(y_t - \bar{y})(y_{t+k} - \bar{y})$

$$\approx \frac{1}{n}\sum_{t=1}^{n-k}(y_t - \bar{y})(y_{t+k} - \bar{y})$$

c_0: Sample variance $\quad c_0 = \frac{1}{n}\sum_t (y_t - \bar{y})^2$

Sample ACF of lag k: $\quad r_k = \frac{c_k}{c_0}; k = 1,2, \ldots$

Empirical Version of Yule-Walker Equations for AR(p) Processes

$$y_t = \mu + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + u_t$$

Yule-Walker equations

$$\rho_1 = \phi_1 \rho_0 + \phi_2 \rho_1 + \cdots + \phi_p \rho_{p-1}$$

$$\vdots$$

$$\rho_p = \phi_1 \rho_{p-1} + \phi_2 \rho_{p-2} + \cdots + \phi_p \rho_0$$

$\hat{\phi}_1, \hat{\phi}_2, \cdots, \hat{\phi}_p$: Estimators of $\phi_1, \phi_2, \cdots\ , \phi_p$

Replacing ρ_k by r_k (γ_k by c_k) and ϕ_k by $\hat{\phi}_k$ leads to empirical version of these equations

$$
\begin{aligned}
r_1 &= \hat{\phi}_1 r_0 + \hat{\phi}_2 r_1 + \cdots + \hat{\phi}_p r_{p-1} \\
&\vdots \\
r_p &= \hat{\phi}_1 r_{p-1} + \hat{\phi}_2 r_{p-2} + \cdots + \hat{\phi}_p r_0
\end{aligned}
$$

or

$$
\left.
\begin{aligned}
c_0 &= \hat{\phi}_1 c_1 + \cdots + \hat{\phi}_p c_p + \hat{\sigma}_u^2 \\
c_1 &= \hat{\phi}_1 c_0 + \cdots + \hat{\phi}_p c_{p-1} \\
&\vdots \\
c_p &= \hat{\phi}_1 c_{p-1} + \cdots + \hat{\phi}_p c_0
\end{aligned}
\right\}
$$

Estimation of parameters for AR(1) Process:

$y_t = \mu + \phi_1 y_{t-1} + u_t$ $\Rightarrow AR(1)$

Method of Moments: Compare parameters by corresponding sample estimates.

Solve empirical Yule-Walker equations for $\phi' s$.

$\hat{\mu}, \hat{\phi}_1, \hat{\sigma}_u^2$: Moment Estimators of $\mu, \phi_{1,}, \sigma_u^2$

For estimating μ

$$\bar{Y} = \hat{\mu} + \hat{\phi}_1 \bar{Y} \Rightarrow \hat{\mu} = (1 - \hat{\phi}_1)\bar{Y}$$

Yule Walker Equations for AR(1)

$$\left.\begin{aligned}\gamma_0 &= \phi_1\gamma_1 + \sigma_u^2\\ \gamma_1 &= \phi_1\gamma_0\end{aligned}\right\}$$

Replacing parameters by estimators

$$c_0 = \hat{\phi}_1 c_1 + \hat{\sigma}_u^2;$$

$$c_1 = \hat{\phi}_1 c_0 \quad \Longrightarrow$$

$$\hat{\sigma}_u^2 = c_0 - r_1 c_1 = c_0(1 - r_1^2) = c_0(1 - \hat{\phi}_1^2)$$

Hence

$$\hat{\phi}_1 = \frac{c_1}{c_0} = r_1, \qquad \hat{\sigma}_u^2 = \left(1 - \hat{\phi}_1^2\right)\mathrm{c}_0$$

Method of Least squares:

$\hat{\mu}, \hat{\phi}_1$: Least square estimators of μ, ϕ_1 minimizing residual SS

$$S = \sum \left(y_t - \hat{\mu} - \hat{\phi}_1 y_{t-1}\right)^2$$

$$c_0 = \frac{1}{n} \sum (y_t - \bar{Y})^2$$

$$= \sum \left[(y_t - \bar{Y}) - \hat{\phi}_1 (y_{t-1} - \bar{Y}) - \{\hat{\mu} - (1 - \hat{\phi}_1)\bar{Y}\}\right]^2$$

$$\sum (y_t - \bar{Y})(y_{t-1} - \bar{Y}) = n c_1$$

$$= n c_0 + n\hat{\phi}_1^2 c_0 - 2n\hat{\phi}_1 c_1 + \text{n}\{\hat{\mu} - (1 - \hat{\phi}_1)\bar{Y}\}^2$$

$$c_1 = r_1 c_0$$

$$= n c_0\left(\hat{\phi}_1^2 - 2\hat{\phi}_1 r_1\right) + n c_0 + \text{n}\{\hat{\mu} - (1 - \hat{\phi}_1)\bar{Y}\}^2$$

$$\hat{\phi}_1^2 - 2\hat{\phi}_1 r_1 + r_1^2 - r_1^2 = (\hat{\phi}_1 - r_1)^2 - r_1^2$$

$$S = nc_0(\hat{\phi}_1^2 - 2\hat{\phi}_1 r_1 + r_1^2 - r_1^2) + nc_0 + \mathrm{n}\{\hat{\mu} - (1 - \hat{\phi}_1)\bar{Y}\}^2$$

$$= \underbrace{nc_0\left(\hat{\phi}_1 - r_1\right)^2}_{\geq 0} + nc_0(1 - r_1^2) + \underbrace{\mathrm{n}\{\hat{\mu} - \left(1 - \hat{\phi}_1\right)\bar{Y}\}^2}_{\geq 0}$$

S is minimum when

$$\hat{\mu} = \left(1 - \hat{\phi}_1\right)\bar{Y}$$

$$\hat{\phi}_1 = \frac{c_1}{c_0} = r_1$$

Alternatively we can minimize $S = \sum(y_t - \hat{\mu} - \hat{\phi}_1 y_{t-1})^2$ using differentiation.

$$\frac{\partial S}{\partial \hat{\mu}} = -2 \sum (y_t - \hat{\mu} - \hat{\phi}_1 y_{t-1}) = 0$$

$$\frac{\partial S}{\partial \hat{\phi}_1} = -2 \sum (y_t - \hat{\mu} - \hat{\phi}_1 y_{t-1}) y_{t-1} = 0$$

Estimation of parameters for AR(2) Process:

$$y_t = \mu + \phi_1 y_{t-1} + \phi_2 y_{t-2} + u_t$$

Method of Moments: Empirical Yule-Walker equations

$$r_1 = \hat{\phi}_1 + \hat{\phi}_2 r_1$$

$$r_1 - r_1 r_2 = \hat{\phi}_1 - r_1^2 \hat{\phi}_1$$

$$r_2 = \hat{\phi}_1 r_1 + \hat{\phi}_2$$

$$\Rightarrow \hat{\phi}_1 = \frac{r_1 - r_1 r_2}{1 - r_1^2};\ \hat{\phi}_2 = \frac{r_2 - r_1^2}{1 - r_1^2}.$$

$$\bar{Y} = \hat{\mu} + \hat{\phi}_1 \bar{Y} + \hat{\phi}_2 \bar{Y} \Rightarrow \hat{\mu} = \bar{Y}(1 - \hat{\phi}_1 - \hat{\phi}_2)$$

Least Squares estimators ≈ Method of moments estimators

Estimation of Parameters of AR(p) process:

$$y_t = \mu + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + u_t \qquad (1)$$

$\hat{\phi}_1, \hat{\phi}_2, \cdots, \hat{\phi}_p$: Estimators of $\phi_1, \phi_2, \cdots, \phi_p$

Empirical version of Yule-Walker equations

$$\begin{gathered} r_1 = \hat{\phi}_1 r_0 + \hat{\phi}_2 r_1 + \cdots + \hat{\phi}_p r_{p-1} \\ \vdots \\ r_p = \hat{\phi}_1 r_{p-1} + \hat{\phi}_2 r_{p-2} + \cdots + \hat{\phi}_p r_0 \end{gathered} \qquad (2)$$

Obtain $\hat{\phi}_1, \hat{\phi}_2, \cdots, \hat{\phi}_p$ by solving these equations.

Estimate μ by

$$\hat{\mu} = \bar{Y}\left(1 - \hat{\phi}_1 - \hat{\phi}_2 - \cdots - \hat{\phi}_p\right). \quad (3)$$

- Moments estimators are very close to least squares estimators.
- If roots of the AR polynomial are close to unit circle, this method will give estimates which are far from actual parameter values.

Estimating parameters of Moving Average Processes

Does method of moments work to fit MA processes?

Ex: MA(1) process

$$y_t = u_t - \theta u_{t-1}, \qquad (1)$$

We have

$$\gamma_0 = \sigma_u^2(1+\theta^2)\,, \qquad \gamma_1 = -\sigma_u^2\theta.$$

Hence,

$$\frac{\gamma_1}{\gamma_0} = -\frac{\theta}{1+\theta^2} \Longrightarrow \gamma_1\theta^2 + \theta\gamma_0 + \gamma_1 = 0$$

Replacing γ_0 and γ_1 by their sample estimates c_0 and c_1 and θ by $\hat{\theta}$, we obtain

$$c_1\hat{\theta}^2 + c_0\hat{\theta} + c_1 = 0. \tag{2}$$

The solution for $\hat{\theta}$ is

$$\hat{\theta} = -\frac{c_0}{2c_1} \pm \sqrt{\left(\frac{c_0}{2c_1}\right)^2 - 1}\,. \tag{3}$$

If one of the two solutions in (3) give invertible process, select that value as an estimate of θ.

If $c_0 = 1,\ c_1 = 0.4,$ the two solutions are $\hat{\theta} = -0.5, -2.$

Process is invertible for $\hat{\theta} = -0.5$, we use it as estimate of θ.

If $c_0 = 1,\ c_1 = 0.5$, then $\hat{\theta} = -1$. Process is not invertible for $\hat{\theta} = -1$.

If $c_0 \leq 2c_1$, we have

$$\hat{\theta} = -\frac{c_0}{2c_1} \pm i\sqrt{1 - \left(\frac{c_0}{2c_1}\right)^2}\ . \qquad (4)$$

Hence $|\hat{\theta}| = 1$ and the process is not invertible.

Fitting MA and mixed ARMA models require numerical optimization techniques.

Estimation of Moving Average Processes using Innovation Algorithm:

Fitted MA(m) model is

$$y_t = u_t + \hat{\theta}_{m1} u_{t-1} + \cdots + \hat{\theta}_{mm} u_{t-m},$$

$\{u_m\}$: White noise process $(0, v_m)$

Innovation estimate of MA parameter

$$\hat{\theta}_{m,m-k} = v_k^{-1}\left[c_{m-k} - \sum_{j=0}^{k-1} \hat{\theta}_{m,m-j}\hat{\theta}_{k,k-j} v_j\right], k = 0, \ldots, m-1$$

$$v_m = c_0 - \sum_{j=0}^{m-1} \hat{\theta}_{m,m-j}^2 v_j$$